Status	Finished
Started	Wednesday, 28 October 2026, 9:29 PM
Completed	Wednesday, 28 October 2026, 9:33 PM
Duration	3 mins 57 secs
Grade	**12.00** out of 15.00 (**80**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Sorting elements in a list
b. Deleting elements from a list
c. Inserting elements into a list
d. Determining whether an element is present in a list ✓

Question **2**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the middle element in the list
b. Compare the target element with the first element in the list
c. Divide the list into two equal parts
d. Sort the list ✗

Question **3**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Sequential search
c. Hash search
d. Binary search ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Binary search
- c. Both (1) & (3)
- d. Hash search

Question **5**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 90 and 100
- b. 94 and 99 ✗
- c. 89 and 94
- d. 90 and 99

Question **6**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the element is found in the list ✓
- b. When the search algorithm finishes
- c. When the list is sorted
- d. When the list contains duplicate elements

Question **7**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the left sublist
c. In the middle of the list
d. In the right sublist ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O🥊
c. O(n log n)
d. O(log n) ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It compares elements sequentially
b. It works on unsorted lists
c. It can be applied only if the list is sorted ✓
d. It always starts from the beginning of the list

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The list is sorted
c. The search continues from the beginning
d. The search continues in the left or right sublist ✓

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

b. Binary search algorithm is not efficient when the data elements more than 1500 ✓

c. Must use a sorted array

d. There must be a mechanism to access middle element directly

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Bubble search

b. Linear search ✓

c. Quick search

d. Merge search

Question **13**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist

b. The search continues in the right sublist

c. The search ends successfully ✓

d. The list is sorted

Question **14**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100

b. 10 ✓

c. 1000

d. 500

Question **15**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element
b. The search stops
c. The lower middle element is chosen as the middle element ✕
d. The middle element is chosen randomly